

Dynamics Test Answers

Ethan Wang

April 27, 2026

Blocks

1. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 144N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:

(a) The acceleration of the system

Solution: The acceleration of the system is -3 ms^{-2}

(b) The acceleration of each block

Solution: Each block has an acceleration of -3 ms^{-2}

(c) The net force on each block

Solution: Block 1 has a net force of -12N
Block 2 has a net force of -72N
Block 3 has a net force of -60N

(d) The contact forces for each block

Solution: Block 1 has 1 force acting on it. They are:
-12N Contact Force from block 2
Block 2 has 2 forces acting on it. They are:
12N Contact Force from block 1, -84N Contact Force from block 3
Block 3 has 2 forces acting on it. They are:
-144N Applied, 84N Contact Force from block 2

2. (6 points) The diagram shows 2 blocks in contact on a smooth surface. A 224N force acts on block 1 from the left, and the surface can be considered frictionless. Calculate:

(a) The acceleration of the system

Solution: The acceleration of the system is 8 ms^{-2}

(b) The acceleration of each block

Solution: Each block has an acceleration of 8 ms^{-2}

(c) The net force on each block

Solution: Block 1 has a net force of 32N
Block 2 has a net force of 192N

- (d) The contact forces for each block

Solution: Block 1 has 2 forces acting on it. They are:
224N Applied, -192N Contact Force from block 2
Block 2 has 1 force acting on it. They are:
192N Contact Force from block 1

3. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 30N force acts on block 1 from the left, and the surface can be considered frictionless. Calculate:

- (a) The acceleration of the system

Solution: The acceleration of the system is 1 ms^{-2}

- (b) The acceleration of each block

Solution: Each block has an acceleration of 1 ms^{-2}

- (c) The net force on each block

Solution: Block 1 has a net force of 1N
Block 2 has a net force of 20N
Block 3 has a net force of 9N

- (d) The contact forces for each block

Solution: Block 1 has 2 forces acting on it. They are:
30N Applied, -29N Contact Force from block 2
Block 2 has 2 forces acting on it. They are:
29N Contact Force from block 1, -9N Contact Force from block 3
Block 3 has 1 force acting on it. They are:
9N Contact Force from block 2

4. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 12N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:

- (a) The acceleration of the system

Solution: The acceleration of the system is -4 ms^{-2}

- (b) The acceleration of each block

Solution: Each block has an acceleration of -4 ms^{-2}

- (c) The net force on each block

Solution: Block 1 has a net force of -4N
Block 2 has a net force of -4N
Block 3 has a net force of -4N

- (d) The contact forces for each block

Solution: Block 1 has 1 force acting on it. They are:
-4N Contact Force from block 2
Block 2 has 2 forces acting on it. They are:
4N Contact Force from block 1, -8N Contact Force from block 3
Block 3 has 2 forces acting on it. They are:
-12N Applied, 8N Contact Force from block 2

5. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 234N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:

- (a) The acceleration of the system

Solution: The acceleration of the system is -9 ms^{-2}

- (b) The acceleration of each block

Solution: Each block has an acceleration of -9 ms^{-2}

- (c) The net force on each block

Solution: Block 1 has a net force of -63N
Block 2 has a net force of -162N
Block 3 has a net force of -9N

- (d) The contact forces for each block

Solution: Block 1 has 1 force acting on it. They are:
-63N Contact Force from block 2
Block 2 has 2 forces acting on it. They are:
63N Contact Force from block 1, -225N Contact Force from block 3
Block 3 has 2 forces acting on it. They are:
-234N Applied, 225N Contact Force from block 2

6. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 423N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:

- (a) The acceleration of the system

Solution: The acceleration of the system is -9 ms^{-2}

- (b) The acceleration of each block

Solution: Each block has an acceleration of -9 ms^{-2}

- (c) The net force on each block

Solution: Block 1 has a net force of -216N
Block 2 has a net force of -72N
Block 3 has a net force of -135N

(d) The contact forces for each block

Solution: Block 1 has 1 force acting on it. They are:
-216N Contact Force from block 2
Block 2 has 2 forces acting on it. They are:
216N Contact Force from block 1, -288N Contact Force from block 3
Block 3 has 2 forces acting on it. They are:
-423N Applied, 288N Contact Force from block 2

7. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 116N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:

(a) The acceleration of the system

Solution: The acceleration of the system is -4 ms^{-2}

(b) The acceleration of each block

Solution: Each block has an acceleration of -4 ms^{-2}

(c) The net force on each block

Solution: Block 1 has a net force of -40N
Block 2 has a net force of -4N
Block 3 has a net force of -72N

(d) The contact forces for each block

Solution: Block 1 has 1 force acting on it. They are:
-40N Contact Force from block 2
Block 2 has 2 forces acting on it. They are:
40N Contact Force from block 1, -44N Contact Force from block 3
Block 3 has 2 forces acting on it. They are:
-116N Applied, 44N Contact Force from block 2

8. (6 points) The diagram shows 2 blocks in contact on a smooth surface. A 126N force acts on block 2 from the right, and the surface can be considered frictionless. Calculate:

(a) The acceleration of the system

Solution: The acceleration of the system is -9 ms^{-2}

(b) The acceleration of each block

Solution: Each block has an acceleration of -9 ms^{-2}

- (c) The net force on each block

Solution: Block 1 has a net force of -90N
Block 2 has a net force of -36N

- (d) The contact forces for each block

Solution: Block 1 has 1 force acting on it. They are:
 -90N Contact Force from block 2
Block 2 has 2 forces acting on it. They are:
 -126N Applied, 90N Contact Force from block 1

9. (6 points) The diagram shows 4 blocks in contact on a smooth surface. A 306N force acts on block 1 from the left, and the surface can be considered frictionless. Calculate:

- (a) The acceleration of the system

Solution: The acceleration of the system is 6 ms^{-2}

- (b) The acceleration of each block

Solution: Each block has an acceleration of 6 ms^{-2}

- (c) The net force on each block

Solution: Block 1 has a net force of 42N
Block 2 has a net force of 60N
Block 3 has a net force of 60N
Block 4 has a net force of 144N

- (d) The contact forces for each block

Solution: Block 1 has 2 forces acting on it. They are:
 306N Applied, -264N Contact Force from block 2
Block 2 has 2 forces acting on it. They are:
 264N Contact Force from block 1, -204N Contact Force from block 3
Block 3 has 2 forces acting on it. They are:
 204N Contact Force from block 2, -144N Contact Force from block 4
Block 4 has 1 force acting on it. They are:
 144N Contact Force from block 3

10. (6 points) The diagram shows 4 blocks in contact on a smooth surface. A 450N force acts on block 1 from the left, and the surface can be considered frictionless. Calculate:

- (a) The acceleration of the system

Solution: The acceleration of the system is 9 ms^{-2}

(b) The acceleration of each block

Solution: Each block has an acceleration of 9 ms^{-2}

(c) The net force on each block

Solution: Block 1 has a net force of 18N

Block 2 has a net force of 216N

Block 3 has a net force of 72N

Block 4 has a net force of 144N

(d) The contact forces for each block

Solution: Block 1 has 2 forces acting on it. They are:

450N Applied, -432N Contact Force from block 2

Block 2 has 2 forces acting on it. They are:

432N Contact Force from block 1, -216N Contact Force from block 3

Block 3 has 2 forces acting on it. They are:

216N Contact Force from block 2, -144N Contact Force from block 4

Block 4 has 1 force acting on it. They are:

144N Contact Force from block 3